

Title : WannaCry Ransomware Attack (2017)

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a)Attack Name :

WannaCry Ransomware Attack 2017.

It is a Worm-based Attack that first appeared on the date 12 May 2017.

It was Crypto-worm malware with self replicative capabilities.

b)Background :

One of the most popular ransomware attacks is WannaCry, which occurred in May 2017. This cyber-attack happened on 12 May 2017 and spread globally very quickly. Within only a couple of days of its operation, it infected over 200,000 computers located in more than 150 countries all over the world. WannaCry infected many sectors, such as healthcare services, manufacturing, financial, transport industries, and even governments. One of the reasons of such a wide spread was the absence of Microsoft's security update in many computers.

The WannaCry ransomware was based on the vulnerability of SMBv1 (Server Message Block version 1) – one of the Windows network protocols. Using the exploit EternalBlue, attackers exploited this vulnerability and got access to computers without updates. The EternalBlue exploit was created by the U.S. National Security Agency (NSA) initially for intelligence purposes. However, in April 2017, the hacking group called Shadow Brokers leaked the exploit and made it accessible to everybody.

In March 2017, the MS17-010 security update was issued by Microsoft. Thus, at the time of the attack, there were thousands of vulnerable computers in the world. Attackers used this fact and launched WannaCry ransomware.

As soon as the computer was infected, the ransomware encrypted all important files and showed the ransom message, requiring the victim to transfer \$300 in Bitcoin to the specified account. In case of the non-payment during the period of three days, the ransom raised to \$600. In addition to the file encryption, WannaCry acts as a worm and looks for other computers in the network with the same vulnerability without the intervention of users. That is the reason of such a rapid spreading through the organization.

The WannaCry attack caused big problems in many countries all over the world. Surgeries were canceled in hospitals, production was stopped in many companies, and businesses had no access to their data. One of the most affected organizations is NHS (National Health Service) of the United Kingdom – doctors could not see the patients' records since they were encrypted.

Currently, WannaCry is used as a practical example to learn about the ransomware attack and about cybersecurity in general. This ransomware showed that the delay of the updates, the use of old protocols, and the lack of backups can lead to catastrophic consequences.

c)Attack type : Worm-based Ransomware

WannaCry is regarded as a worm-ransomware since it does not only encrypt the files of the victim and request a ransom but also uses the EternalBlue exploit to propagate itself to other computers via the SMBv1 Protocol (Port 445). It must be emphasized that unlike the ransomware that usually spreads via malicious emails, WannaCry could infect other computers without any involvement of users.

d)TimeLine of Event :

Date	Events
Before 2017	Microsoft Windows SMBv1 protocol had a security flaw that could enable remote code execution.
Before 2017	NSA found out about this vulnerability and created a hack code named EternalBlue. The agency also created another backdoor tool called DoublePulsar for use after the hack.
14 March 2017	Microsoft came up with MS17-010 security patch that would resolve the vulnerability in SMBv1 protocol.
14 April 2017	The hacker collective called Shadow Brokers disclosed EternalBlue and other cyberweapons of NSA.
12 May 2017	WannaCry ransomware attack was launched targeting unpatched computers of Microsoft using EternalBlue exploit.
12-13 May 2017	The WannaCry worm spread widely by exploiting vulnerabilities of SMBv1 over port 445 on many computers. More than 200,000 computers have been affected in over 150 countries.
13 May 2017	Security analyst Marcus Hutchins registered WannaCry Kill Switch that stopped the spreading of this malware.

Subsequent Days Microsoft rolled out emergency patches for unsupported Windows operating systems, and businesses all around the world have begun to patch their systems, disabled SMBv1 and quarantined infected devices.

Post Outbreak WannaCry has become one of the most researched ransomware cases.

e) Attack Method :

The WannaCry ransomware infected Windows operating system which did not install Microsoft patches by targeting the vulnerability in the Server Message Block Version 1 (SMBv1). The EternalBlue exploit was used in the attack to access vulnerable machines via TCP Port 445. Once the machine is exploited, the malware executes its malicious code and encrypts all the files present and displays a ransom message requesting money from the victim in the form of Bitcoins.

The difference between WannaCry ransomware and conventional ransomware is that the WannaCry ransomware has the ability to act like a worm. The malware would self-replicate once a system was infected and automatically scan other machines on the local network which were running the vulnerable SMBv1 service.

f) Impact :

Within a few days, 200,000 computers in 150+ countries were infected, making WannaCry one of the biggest ransomware attacks.

There were critical infrastructures that were severely impacted by this malware. These include the health sector, manufacturing industry, transport, finance, and government.

National Health Service of the UK was greatly impacted as it could not perform surgeries and appointments due to inaccessibility of patients' information.

Organizations were forced to halt their operations for some time due to the financial impact and inefficiency.

Key data files were encrypted by the malware, and access to them could not be gained without a decryption key.

This malware could spread itself across the network via vulnerable systems using SMBv1.

The need for proper maintenance of systems, regular backup of data, segmentation of networks, and vulnerability management came into focus during the incident.

g) Response & Recovery :

The infected computers were disconnected from the network so that no further spread of the ransomware to other computers would take place.

The MS17-010 patch was implemented, and the use of SMBv1 was terminated.

A firewall was set up to block SMB communication (Port 445), when it was not needed.

Malware on the infected machines was detected and removed, and files affected by the ransomware were recovered from backup whenever possible.

Security researcher Marcus Hutchins registered the WannaCry Kill Switch domain, greatly reducing the rate of infection.

Microsoft issued emergency security patches for some outdated Windows versions to mitigate the vulnerabilities.

Many companies adopted new practices in patch management, backing up data regularly, implementing network segmentation, and increasing employee awareness after the incident.

h) Prevention Measures :

Block SMBv1 and instead use SMBv2 or above whenever it is able to.

If SMB access is not needed, port 445 should be blocked or restricted by a firewall.

Network Segmentation: Deploy tools that will prevent a malware from traversing throughout the network.

Remember to make frequent offline backups and try to restore the data after an attack.

Deploy antivirus/EDR solutions to catch and block ransomware functionalities.

Perform vulnerability assessments regularly & remediate identified vulnerabilities and weaknesses.

Train employees to identify phishing emails and other forms of social engineering attacks that are often used to deliver ransomware.

Develop and periodically exercise the IRP (Incident Response Plan) ensuring that you have some sort of a plan to deal with security events in case the need arise.

This means continuous monitoring of network and endpoint activities, through SIEM, IDS/IPS, or EDR to detect signs of suspicious behavior as early as possible.

I) Conclusion :

The WannaCry ransomware attack showed how a single unpatched vulnerability can lead to a large-scale cyberattack. It affected thousands of organizations worldwide and caused significant

financial and operational damage. The incident highlighted the importance of timely patch management, regular backups, network security, and a well-prepared incident response plan. Even today, WannaCry remains an important case study for understanding ransomware attacks and improving cybersecurity practices.

J) References :

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